

SAEFormer: Semantic-Aware Enhancement for Temporal Action Localization

Abstract

Temporal action localization aims to identify action instances in untrimmed videos and predict their temporal boundaries. Existing Transformer-based methods improve temporal dependency modeling, but stacked attention aggregation may smooth short-range variations and weaken discriminative cues near ambiguous action boundaries. To address this problem, we propose SAEFormer, a representation-level cue enhancement framework for multi-scale temporal action localization. In this paper, the core of SAEFormer is SAEBlock, which consists of Learnable Multi-branch Semantic Fusion (LMSF) and Boundary-aware Enhancement (BAE). LMSF mitigates the smoothing effect on local discriminative cues around attention aggregation while preserving contextual information. BAE enhances transition-sensitive temporal representation cues by residually reweighting boundary-related temporal positions, and learns boundary-oriented reweighting without introducing an additional boundary-level loss.

Experiments on three challenging benchmarks validate the competitiveness of SAEFormer. It achieves 69.7% and 71.4% average mAP on THUMOS14 under two feature settings, 36.7% on ActivityNet-1.3 under the main setting, and 26.5% / 23.7% for EPIC-

KITCHENS-100 verb / noun localization. Ablation studies further show that LMSF and BAE provide complementary gains.

Keywords: Temporal action localization, Untrimmed video understanding, Semantic-aware enhancement, Boundary-oriented reweighting

1 Introduction

Temporal action localization (TAL) is an important task in long-form intelligent video understanding. It aims to identify action instances in untrimmed videos and estimate their start and end timestamps [1,2], supporting applications such as video content analysis, multimedia retrieval, sports analysis, and intelligent surveillance [3-5]. In these scenarios, TAL converts weakly structured video streams into temporally localized action segments, providing structured evidence for retrieval, monitoring, and downstream decision-making. Unlike trimmed action recognition, TAL must reason over long temporal sequences that usually involve large variations in action duration, ambiguous action-background transitions [6], and a high proportion of background content. Therefore, TAL is sensitive not only to semantic recognition errors but also to small boundary shifts, because such shifts affect detection quality and downstream decision support under strict temporal intersection-over-union thresholds.

Existing TAL methods have evolved from convolutional and graph-structured architectures to Transformer-based frameworks [7-11]. Convolutional models provide

useful local temporal inductive biases, whereas graph-structured designs model relations among clips or proposals. Transformer-based methods further apply attention along the temporal dimension to improve long-range temporal dependency modeling. In particular, ActionFormer [10] establishes a strong point-based baseline by combining local self-attention with multi-scale feature pyramids. This line of work has made attention-based temporal modeling a widely adopted direction in one-stage TAL. However, when multiple modules are stacked, attention-based context aggregation may also reduce local temporal contrast. It is therefore necessary to perform representation-level cue enhancement to complement the attention mechanism rather than replace the detection framework.

This problem is particularly prominent in background-dominated videos. Attention aggregation helps exchange contextual information, but it may smooth short-range temporal variations and weaken differences between adjacent action and background clips. When multiple temporal modeling modules are stacked, this effect can be further amplified. Related studies on vision Transformers also suggest that self-attention may cause patch or token representations to converge or become over-smoothed [12,13]. In long-sequence TAL modeling, a similar tendency may further weaken boundary-sensitive local variations. As a result, a model may capture global context while still losing part of the local evidence needed to distinguish action starts, action interiors, and action ends.

This observation raises two representation-level questions: how to preserve local discriminative cues during attention-based context aggregation, and how to enhance

boundary-oriented temporal transitions without introducing additional boundary supervision. We argue that effective temporal localization requires representation-level cues with different roles. Local saliency and identity information help preserve short-range temporal contrast, convolutional modulation provides local context, and boundary-oriented temporal responses emphasize transition-sensitive positions. Therefore, we aim to organize these cues inside the temporal backbone, without replacing the attention-based backbone modeling paradigm or relying on additional boundary supervision.

To address this problem, we propose SAEFormer, a one-stage, multi-scale, point-based framework for temporal action localization. In this paper, semantic-aware enhancement refers to representation-level cue enhancement rather than external semantic reasoning. SAEBlock preserves local discriminative cues before and after attention aggregation through lightweight learnable cue fusion. BAE learns boundary-oriented temporal reweighting through the standard temporal action localization objective, without adding boundary-level supervision.

The main contributions of this paper are summarized as follows:

We propose SAEBlock, in which LMSF mitigates the smoothing effect of stacked temporal attention on local discriminative cues during context aggregation, thereby improving the discriminability of temporal representations near boundaries.

We introduce BAE, which enhances transition-sensitive temporal representation cues by residually reweighting boundary-related temporal positions, without requiring additional boundary supervision.

We conduct extensive experiments on multiple mainstream datasets and achieve competitive performance, while systematic ablations on THUMOS14 validate the effectiveness of the proposed modules.

2 Related Work

2.1 Temporal Action Localization (TAL)

Temporal action localization (TAL) aims to locate and classify action instances in untrimmed videos. Existing TAL methods are commonly divided into two-stage and one-stage frameworks [14]. Two-stage methods such as BMN [15] and G-TAD [16] follow a proposal-then-classify paradigm, where candidate temporal segments are first generated and then classified. This design benefits from explicit proposal modeling, but the separation between proposal generation and classification may introduce error propagation and increase inference latency.

One-stage methods reduce pipeline complexity by directly predicting temporal boundaries and action categories. CNN-based methods [17-19] improve end-to-end detection efficiency, while anchor-free methods such as AFSD [8] and DyFADet [20] further remove hand-crafted anchor design and shift toward direct boundary regression. More recent Transformer-based baselines, including RTD-Net [21], ActionFormer [10],

and TallFormer [22], strengthen long-range temporal dependencies and have become strong reference frameworks. These developments reduce pipeline complexity, but long videos with variable action durations and background-dominated content still place high demands on reliable point-level classification and boundary regression.

2.2 Temporal Context Modeling

Effective temporal context modeling is essential for TAL because action instances vary substantially in duration and are often surrounded by background segments. Previous studies have explored convolution-based, graph-structured, and Transformer-based strategies [9,10,23]. CNN-based methods such as TCANet [17] and ContextLoc [18] aggregate multi-granularity temporal contexts, whereas graph-structured designs such as G-TAD [16] model relations among clips or proposals. These designs improve temporal dependency modeling, but their ability to jointly represent local boundary variations and long-range context remains limited in complex untrimmed videos.

Recent studies have therefore introduced more adaptive or boundary-sensitive temporal modeling mechanisms. TriDet [24] replaces self-attention with convolutions adapted to different temporal scales, modeling local context at multiple granularities for point-based detection. TemporalMaxer [25] revisits max-pooling-based context aggregation, and CTAN [26] combines convolution with Transformer-style modeling to preserve fine-grained local variations. DyFADet [20] adaptively adjusts receptive fields and convolutional weights for actions of different durations, while a diffusion-based method [27] introduces generative

denoising diffusion to improve temporal action detection. Related methods such as ASL [28], TE-TAD [29], DualH [19], and BRTAL [30] further emphasize the importance of action-sensitive representation learning, time-aligned detection, hierarchical temporal modeling, and boundary refinement. These studies mainly modify temporal operators, detection heads, or refinement mechanisms. How to preserve cue diversity inside attention blocks remains less directly discussed.

2.3 Transformer-Based TAL Methods

Transformers have achieved strong performance in sequence modeling [23] and have been widely adopted in temporal action localization. ActionFormer [10] integrates local self-attention with a feature pyramid and establishes a strong point-based Transformer baseline. TallFormer [22] introduces a long-memory mechanism to reduce the cost of processing high-resolution long videos. TeST [31] adopts a temporal-spatial separated Transformer architecture to jointly model spatial and temporal cues, and DiGIT [32] advances query-based Transformer detection through a multi-dilated gated encoder and a central-adjacent decoder. These methods mainly improve temporal dependency modeling and decoding strategies over long video sequences.

Other studies extend representation learning and training paradigms for Transformer-based temporal action localization from complementary perspectives. ADS [11] introduces a dual selective token-mixing mechanism to suppress redundant information and enhance discriminative boundary representations. TFFormer [33] performs cross time-frequency

interaction to capture local temporal details and global frequency patterns. AdaTAD [34] explores end-to-end temporal action localization based on large-scale VideoMAE, and Dual DETRs [35] use instance-level and boundary-level queries for multi-label TAD. In parallel with these Transformer-based directions, State-Sensitive Mamba [36] advances temporal localization from the perspective of state-space long-sequence modeling. Different from methods that mainly improve temporal operators, query decoders, or training paradigms, SAEFormer starts from representation enhancement inside the Transformer backbone. It uses LMSF to mitigate the smoothing effect of attention aggregation on local discriminative cues and BAE to perform boundary-oriented reweighting on boundary-related temporal positions. In the full configuration, these two modules jointly form SAEBlock and improve the discriminability of representations near boundaries.

3 Preliminaries

Given an input video, a sequence of clip-level features is first extracted by a pre-trained video feature extractor, such as I3D [37] or SlowFast [38]. The extracted feature

sequence is denoted as $F = \{f_t\}_{t=1}^T$, where T denotes the number of temporal clips and

f_t denotes the feature vector of the t -th clip. The goal of temporal action localization is

to predict a set of action instances $Y = \{y_1, y_2, \dots, y_N\}$ from the extracted feature sequence

F , where N denotes the number of action instances. The n -th ($n = 1, \dots, N$) element in

Y is defined as $y_n = (s_n, e_n, a_n)$, where $s_n \in [1, T]$ and $e_n \in [1, T]$ denote the start and

end temporal indices, respectively, $a_n \in \{1, \dots, N_C\}$ denotes the category label, and N_C

denotes the total number of predefined action categories.

Following the point-based TAL paradigm, the feature sequence can be represented by a set of temporal points, where each point is responsible for action classification and

boundary regression. For a temporal point p_t located inside an action instance

$y_n = (s_n, e_n, a_n)$, the boundary regression targets are defined as follows.:

$$d_t^s = p_t - s_n, \quad d_t^e = e_n - p_t$$

where d_t^s and d_t^e denote the distances from p_t to the start and end boundaries,

respectively. To accommodate action instances with different temporal durations, point-based TAL methods commonly perform prediction on multi-scale temporal features, where

high-resolution levels are used for short actions and low-resolution levels are used for long actions.

4 Methodology

As shown in Fig. 1, SAEFormer takes pre-extracted clip-level features as input. After embedding projection, the features enter a temporal encoding backbone stacked with SAEBlocks. In this paper, "semantic" specifically refers to action-related discriminative representation cues encoded by pre-trained video features, including action-background discrimination cues, local temporal variation cues, and boundary transition cues. It does not refer to external language semantics, category-name semantics, or knowledge-based reasoning semantics. In the full configuration, SAEBlock consists of Learnable Multi-branch Semantic Fusion (LMSF) and BAE. The two modules can also be ablated as separable components, where LMSF is responsible for local cue preservation and BAE for boundary-oriented reweighting. The multi-scale features are then used by a 1D feature pyramid and point-based classification and boundary decoding heads to generate predictions, followed by Soft-NMS for redundancy removal.

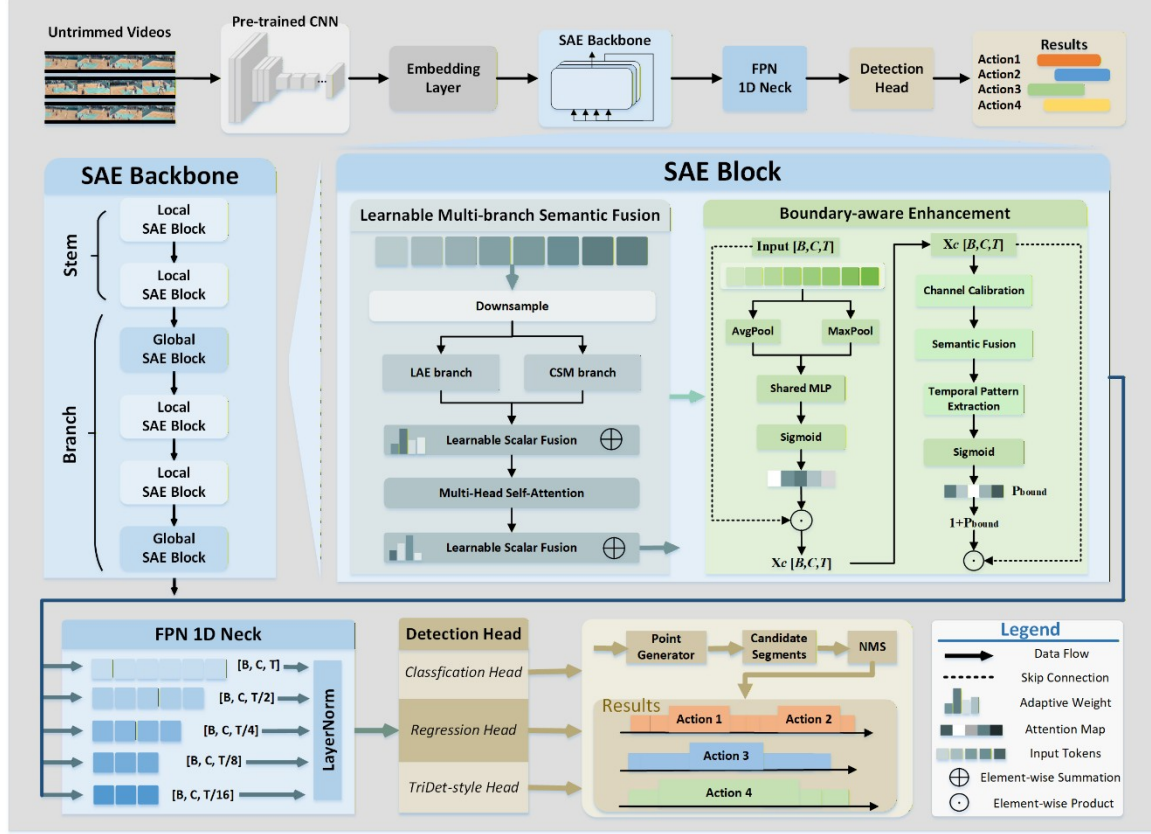


Fig. 1. Overall architecture of SAEFormer. SAEFormer first maps pre-extracted clip-level features into a unified embedding space, and then generates multi-scale temporal features through a temporal encoding backbone composed of stacked SAEBlocks. A full SAEBlock consists of Learnable Multi-branch Semantic Fusion (LMSF) and Boundary-aware Enhancement (BAE). In the ablation experiments, the two modules are enabled separately as separable components. The multi-scale features are fed into a 1D feature pyramid and point-based classification and boundary decoding heads to predict action categories and temporal boundaries.

4.1 Learnable Multi-branch Semantic Fusion (LMSF)

As shown in Fig. 2, LMSF contains Convolutional Semantic Modulation (CSM), an identity reference, and Local Activation Enhancement (LAE). It performs two-stage fusion around Multi-Head Self-Attention (MHSA) to preserve local discriminative cues during attention aggregation.

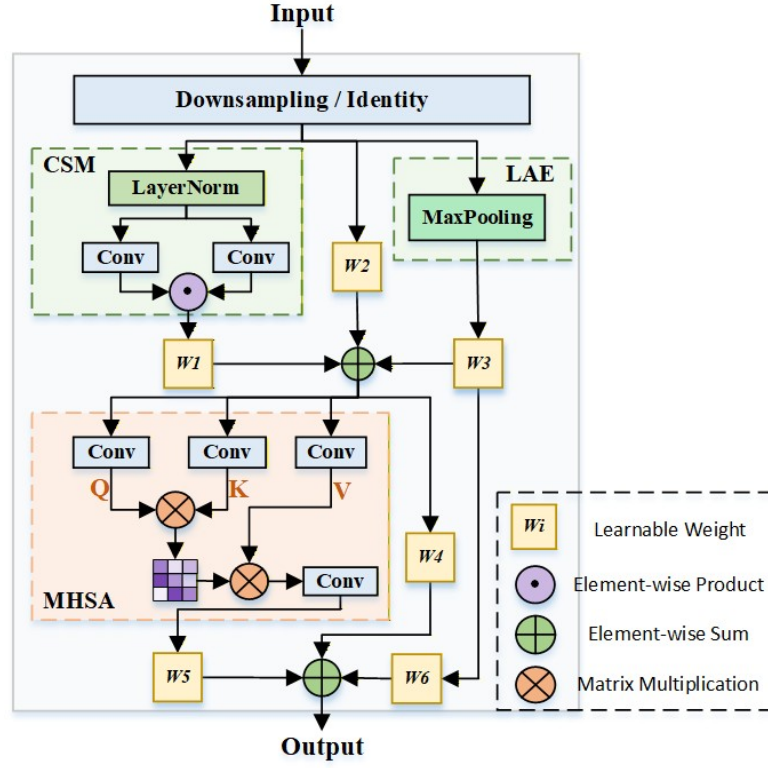


Fig. 2. Schematic diagram of LMSF.

Multi-branch feature extraction. CSM obtains convolutionally modulated features, the identity branch preserves the downsampled input, and LAE preserves short-range peak responses through temporal max pooling, serving as a complementary cue for local discrimination and boundary variation.

First-stage Fusion. The above three branches are fused using learnable scalar weights,

which are optimized end-to-end. Let $\mathbf{F}_{\text{csm}}$, $\mathbf{F}_{\text{id}}$ and $\mathbf{F}_{\text{lae}}$ denote the semantic modulation feature, identity feature, and local saliency feature, respectively. The first-stage fusion is formulated as:

$$\mathbf{F}_{\text{pre}} = w_1 \mathbf{F}_{\text{csm}} + w_2 \mathbf{F}_{\text{id}} + w_3 \mathbf{F}_{\text{lae}}$$

where w_1 , w_2 and w_3 are learnable scalar weights in the current SAEBlock. We adopt simple element-wise weighted summation for feature fusion, which preserves the channel dimension and avoids the additional projection layer required by concatenation-based fusion.

Convolutional Temporal Attention. The fused representation $\mathbf{F}_{\text{pre}}$ is then refined by a convolutional temporal attention module. Instead of directly using linear projections to generate query, key, and value features, we adopt lightweight convolutional projections:

$$\mathbf{Q} = \text{Conv}_q(\mathbf{F}_{\text{pre}}), \mathbf{K} = \text{Conv}_k(\mathbf{F}_{\text{pre}}), \mathbf{V} = \text{Conv}_v(\mathbf{F}_{\text{pre}})$$

Here, $\text{Conv}(\cdot)$ denotes a depth-wise convolution. This design introduces local temporal inductive bias before attention computation while keeping the projection lightweight. For the j -th attention head, the attention output is computed as:

$$\mathbf{F}_{\text{attn}}^j = \text{Softmax}\left(\frac{\mathbf{Q}_j \mathbf{K}_j^\diamond}{\sqrt{D_h}}\right) \mathbf{V}_j$$

where $D_h = C / N_h$, N_h is the number of attention heads, and $\mathbf{Q}_j$, $\mathbf{K}_j$ and $\mathbf{V}_j$ denote

the query, key, and value features of the j -th head. In implementation, the valid temporal

mask is applied in the attention computation to prevent padded positions from participating in feature aggregation.

The outputs of all heads are concatenated and projected to obtain the final attention-enhanced representation:

$$\mathbf{F}_{\text{attn}} = \text{Conv}_{\text{out}} \left(\text{Concat} \left(\mathbf{F}_{\text{attn}}^1, \dots, \mathbf{F}_{\text{attn}}^{N_h} \right) \right)$$

According to the configured attention pattern, this module can be instantiated as either global masked temporal attention or local window attention.

Second-stage Fusion. To further integrate the information after attention, a second learnable scalar fusion is applied among the local saliency feature, the first-stage fused representation, and the attention-enhanced feature:

$$\mathbf{F}_{\text{out}} = w_4 \mathbf{F}_{\text{lae}} + w_5 \mathbf{F}_{\text{attn}} + w_6 \mathbf{F}_{\text{pre}}$$

where w_4, w_5, w_6 are another three learnable scalar weights.

Re-incorporating $\mathbf{F}_{\text{lae}}$ at this stage provides an additional local reference after attention, which may help retain locally peaked responses and boundary-sensitive temporal cues during feature aggregation.

Residual Connection. The output of LMSF is passed through the residual path of the SAEBlock. Specifically, the fused representation is first scaled by a learnable affine parameter before being added back to the residual input through stochastic depth:

$$\mathbf{Y} = \mathbf{X}_{\text{res}} + \text{DropPath}(\gamma \mathbf{F}_{\text{out}})$$

where $\mathbf{X}_{\text{res}}$ denotes the residual input after the optional downsampling operation, and γ denotes the learnable scale parameter. The scale parameter is initialized with a small value to stabilize early training by suppressing the residual branch.

The six fusion weights in LMSF are learned independently in each SAEBlock. They are scalar parameters rather than channel-level, temporal-level, or sample-level gates, and they are not normalized by softmax. Therefore, LMSF should be understood as block-wise learnable scalar cue fusion rather than an input-dependent routing mechanism. For the fusion weights of LMSF, each SAEBlock introduces only six additional block-wise scalar fusion parameters.

4.2 Boundary-aware Enhancement

BAE is inspired by channel recalibration [39], as shown in Fig. 3, channel recalibration highlights informative channels, and subsequent learnable temporal transition responses form boundary-related reweighting. Action boundaries usually correspond to transition regions from background to action or from action to background. BAE learns this reweighting response to enhance boundary-related positions.

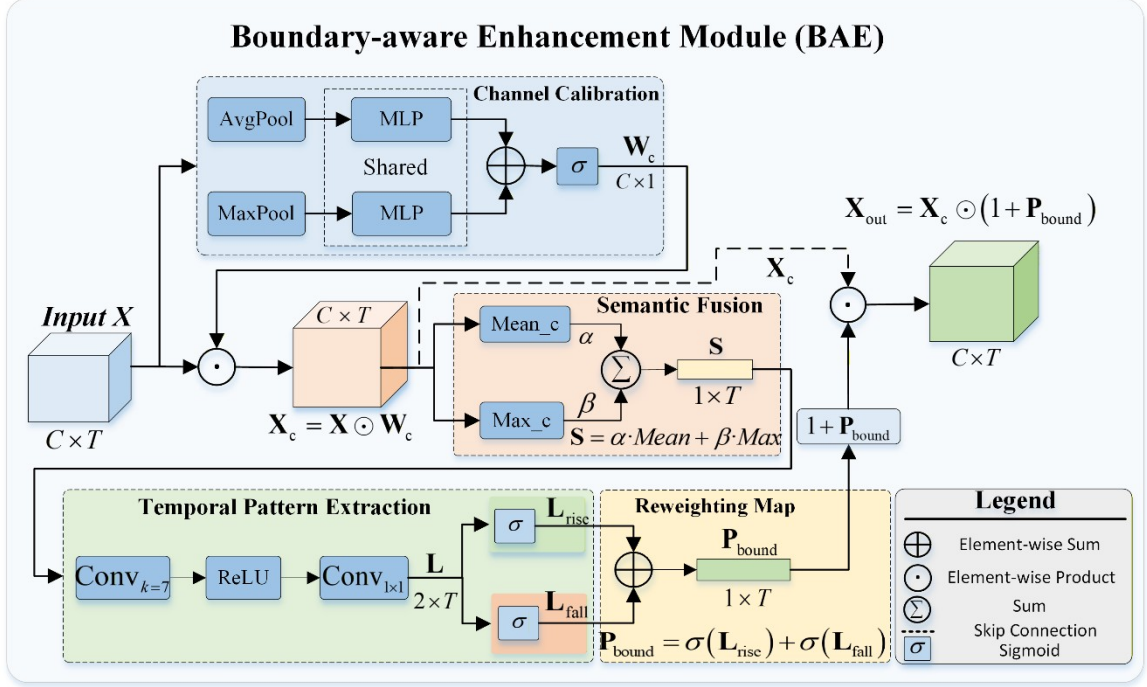


Fig. 3. Schematic diagram of BAE.

The channel calibration weight is computed as:

$$\mathbf{W}_c = \sigma(\mathcal{M}(\text{AvgPool}(\mathbf{X})) + \mathcal{M}(\text{MaxPool}(\mathbf{X})))$$

where $\mathcal{M}$ denotes a shared MLP implemented by two 1×1 convolutions, and σ is the sigmoid activation function. The calibrated feature is then obtained as:

$$\mathbf{X}_c = \mathbf{X} \odot \mathbf{W}_c$$

where $\odot$ denotes element-wise multiplication. This calibration step encourages the module to focus on channels that are more relevant to temporal action patterns before modeling boundary-related variations.

Then, a compact one-dimensional semantic sequence is constructed by computing the average and peak responses along the channel dimension:

$$\mathbf{s}_{\text{avg}} = \text{Mean}_c(\mathbf{X}_c), \mathbf{s}_{\text{max}} = \text{Max}_c(\mathbf{X}_c)$$

where $\mathbf{s}_{\text{avg}}$ and $\mathbf{s}_{\text{max}}$ are in $\mathbb{R}^{1 \times T}$, these two responses provide complementary temporal descriptions. The average response summarizes the overall temporal semantics, while the peak response preserves locally salient activations. To balance these two types of temporal cues, BAE combines them using learnable semantic mixing weights:

$$\mathbf{s} = \alpha \cdot \mathbf{s}_{\text{avg}} + \beta \cdot \mathbf{s}_{\text{max}}, [\alpha, \beta] = \text{Softmax}\left(\begin{bmatrix} a_{\text{avg}} \\ a_{\text{max}} \end{bmatrix}\right)$$

where a_{avg} and a_{max} are two learnable scalar logits used to generate the mixing weights for the average and peak semantic responses. The mixed semantic sequence $\mathbf{s}$ is then fed into a lightweight temporal pattern extraction branch:

$$[\mathbf{L}_{\text{rise}}, \mathbf{L}_{\text{fall}}] = \text{Conv}_{1 \times 1}(\text{ReLU}(\text{Conv}_{k \times 1}(\mathbf{s})))$$

where ReLU denotes the nonlinear activation function. The two output channels $\mathbf{L}_{\text{rise}}$ and

$\mathbf{L}_{\text{fall}}$ describe rising and falling temporal transition responses, respectively. These responses may provide useful cues for action starts and ends, but they are not supervised by explicit boundary labels. Instead, they are learned implicitly through the final TAL objective. The boundary-aware temporal response is computed as:

$$\mathbf{P}_{\text{bound}} = \sigma(\mathbf{L}_{\text{rise}}) + \sigma(\mathbf{L}_{\text{fall}})$$

where $\mathbf{P}_{\text{bound}} \in \mathbb{R}^{1 \times T}$. We use additive fusion rather than competitive normalization because rising and falling responses may provide complementary information for temporal transitions. Since $\mathbf{P}_{\text{bound}}$ serves as a temporal reweighting map rather than a normalized probability distribution, both responses are allowed to contribute jointly. Finally, BAE enhances the calibrated feature through a residual boundary-aware weight:

$$\mathbf{X}_{\text{out}} = \mathbf{X}_{\text{c}} \odot (1 + \mathbf{P}_{\text{bound}})$$

where $\mathbf{P}_{\text{bound}}$ is broadcast along the channel dimension. The residual form preserves the calibrated feature while increasing the responses at temporal positions with stronger boundary-related cues. Compared with fixed first-order difference operators, the learnable rising/falling branch provides a data-adaptive way to model boundary-sensitive temporal variations. The effectiveness of this design is evaluated in the ablation study by comparing it with the fixed-difference variant.

4.3 Point-Based Classification and Boundary Decoding

After the SAEBackbone, the enhanced temporal features are arranged into a multi-scale temporal feature pyramid. Each pyramid level corresponds to a different temporal

stride, allowing short and long action instances to be handled at different temporal resolutions. Let the feature pyramid be denoted as

$$\mathcal{F} = \{\mathbf{F}_l\}_{l=0}^{L-1}$$

where $\mathbf{F}_l$ denotes the temporal feature at the l -th pyramid level and L is the number of pyramid levels. Following the point-based TAL paradigm, a classification branch predicts action category logits at each temporal point, while a regression branch estimates the distances to the start and end boundaries:

$$\mathbf{q}_{l,t} = \mathcal{H}_{cls}(\mathbf{F}_{l,t}), \quad (d_{l,t}^s, d_{l,t}^e) = \mathcal{H}_{reg}(\mathbf{F}_{l,t})$$

where $\mathbf{q}_{l,t}$ denotes the classification logits at the t -th temporal point of the l -th pyramid level, and $d_{l,t}^s$ and $d_{l,t}^e$ denote the predicted distances to the start and end boundaries, respectively.

In the main configuration, the regression branch adopts a Trident-style [24] boundary decoding design. Instead of relying only on direct offset regression, it further uses center offset logits and start/end boundary neighbor logits to estimate temporal offsets. The final left and right offsets are obtained from the predicted boundary distributions and are used to decode the action segment as:

$$\hat{s}_{l,t} = t - d_{l,t}^s, \quad \hat{e}_{l,t} = t + d_{l,t}^e$$

Following the point-based TAL paradigm, the detection head serves as a standard prediction interface. It converts the multi-scale temporal representations produced by SAEBackbone into category scores and temporal boundary offsets.

4.4 Training and Inference

Following standard point-based temporal action localization methods [10,24], temporal points are assigned labels during training according to ground-truth segments, regression ranges, and center-sampling constraints. We use sigmoid focal loss for classification and one-dimensional DIoU loss for boundary regression on positive samples:

$$L = L_{focal} + \lambda_{reg} L_{DIoU}$$

This objective combines the classification loss and boundary regression loss with a balancing coefficient λ_{reg} . BAE does not introduce an additional boundary-level loss. Its boundary-oriented temporal reweighting is learned indirectly through the final temporal action localization objective.

During inference, SAEFormer filters low-confidence temporal points, decodes top-ranked candidate points into action segments, and applies Soft-NMS to remove redundant predictions:

$$\hat{Y} = SoftNMS(\{\hat{s}_i, \hat{e}_i, \hat{a}_i, \hat{q}_i\}_{i=1}^K)$$

Each output contains the predicted start time, end time, category, and confidence score.

5 Experiments

5.1 Experimental Setup

5.1.1 Datasets

We evaluate the proposed SAEFormer on three challenging benchmarks: THUMOS14 [40], ActivityNet-1.3 [1], and EPIC-KITCHENS-100 [41]. THUMOS14 contains 413 untrimmed videos covering 20 action categories. It is divided into validation and test subsets, with 200 videos for validation and 213 videos for testing. ActivityNet-1.3 is a large-scale action dataset containing 200 activity classes and approximately 20,000 videos, with a total duration of more than 600 hours. It is divided into three subsets: 10,024 training videos, 4,926 validation videos, and 5,044 test videos. EPIC-KITCHENS-100 is a large-scale egocentric action dataset. It contains 100 hours of video from 700 sessions, recording cooking activities in different kitchens.

5.1.2 Evaluation Metrics

To evaluate the effectiveness of the proposed method, we follow common evaluation protocols in temporal action localization [10,22,24]. Performance is measured by mean average precision (mAP) under multiple temporal Intersection over Union (tIoU) thresholds. The tIoU thresholds are set to [0.3: 0.1: 0.7] for THUMOS14, [0.5: 0.05: 0.95] for ActivityNet-1.3, and [0.1: 0.1: 0.5] for EPIC-KITCHENS-100. The notation [start: step: end] denotes a set of thresholds that starts from start, uses step as the interval, and ends at

end. For example, [0.3: 0.1: 0.7] corresponds to 0.3, 0.4, 0.5, 0.6, and 0.7. The other threshold settings follow the same notation rule.

5.1.3 Implementation Details

All experiments use pre-extracted clip-level features: THUMOS14 uses I3D [37] and VideoMAE V2 [42], ActivityNet-1.3 reports results with both I3D and TSP [43] features, and EPIC-KITCHENS-100 uses SlowFast [38]. Unless otherwise stated, the ActivityNet implementation details refer to the main TSP setting. The experiments are implemented in Python 3.10 with PyTorch 2.10.0 and CUDA 12.8, and are run on two NVIDIA RTX 3090 GPUs. We train for 54, 50, 17, 30, and 33 epochs on THUMOS14-I3D, THUMOS14-VideoMAE V2, ActivityNet-1.3, EPIC-KITCHENS-100 noun, and EPIC-KITCHENS-100 verb, respectively, with 24, 20, 5, 7, and 10 warm-up epochs. The optimizer uses cosine annealing [44] together with AdamW [45]. The learning rate is $1e-4$ for THUMOS14 and EPIC-KITCHENS-100, and $1e-3$ for ActivityNet-1.3. The batch sizes for THUMOS14-I3D, THUMOS14-VideoMAE V2, ActivityNet-1.3, EPIC-KITCHENS-100 noun, and EPIC-KITCHENS-100 verb are 2, 2, 16, 2, and 2, respectively.

In addition to the above settings, we follow point-level label assignment, center sampling, regression range settings, candidate filtering, and Soft-NMS post-processing protocols consistent with the ActionFormer-style baseline. Because different datasets and feature settings have different temporal resolutions and action-duration distributions, the

local window, MLP dimension, number of candidates, and NMS parameters are set according to the corresponding dataset configuration. Within the same dataset, the main model and ablation experiments use consistent training, inference, and evaluation protocols.

5.2 Comparison Results on Public Datasets

Tables 1-3 compare SAEFormer with state-of-the-art temporal action localization methods on THUMOS14, ActivityNet-1.3, and EPIC-KITCHENS-100. For transparency, each table reports the video features or feature extractors used by comparison methods when available, and includes publicly reported results under the corresponding evaluation protocols. Because different results use different pre-trained features, these results should be interpreted as comparisons based on public reports rather than strictly controlled comparisons under a unified feature setting.

5.2.1 THUMOS14

On THUMOS14, SAEFormer achieves average mAP scores of 69.7% and 71.4% under the I3D and VideoMAE V2 feature settings, respectively, obtaining the best average results among the publicly reported results listed in Table 1. Specifically, under the I3D setting, SAEFormer obtains a small advantage over recent methods. Under the VideoMAE V2 setting, SAEFormer achieves strong results in average mAP and at most tIoU thresholds, but it does not surpass existing methods at every threshold.

For the VideoMAE V2 comparison, we use the VideoMAE V2 model of Wang et al. [42] to extract features and report it separately from I3D-based comparisons to avoid direct

confusion between different feature settings. Because the number of publicly reported same-feature VideoMAE V2 results is still limited, this setting is mainly used to show the performance of SAEFormer when combined with stronger video representations, rather than to provide a complete ranking of all methods under unified features.

Table 1.

Comparison with existing temporal action localization methods on the THUMOS14 dataset. The best results are highlighted in bold.

Method	Venue	Backbone	tIoU					Avg
			@0.	@0.	@0.	@0.	@0.	
			3	4	5	6	7	
ActionFormer [10]	ECCV 2022	I3D	82.1	77.8	71	59.4	43.9	66.8
		VideoMAE V2	84	79.6	73	63.5	47.7	69.6
TemporalMax er [25]	arXiv 2023	I3D	82.8	78.9	71.8	60.5	44.7	67.7
TriDet [24]	CVPR 2023	I3D	83.6	80.1	72.9	62.4	47.4	69.3
TFFormer [33]	T-CSVT 2024	I3D	82.1	78.9	72	60.8	44.9	67.8
ASL [28]	ICCV 2023	I3D	83.1	79.0	71.7	59.7	45.8	67.9
TE-TAD [29]	CVPR 2024	I3D	83.3	78.4	71.3	60.7	45.6	67.9
DualH [19]	FG 2024	I3D	83.6	79.5	72.2	60	44.9	68
CTAN [26]	KBS 2024	I3D	82.9	79.7	72.2	61.2	45.7	68.3
		VideoMAE V2	84.5	80.6	74.6	61.8	48.9	70.1
DyFADet [20]	ECCV 2024	I3D	84	80.1	72.7	61.1	47.9	69.2
GDD-TAD [27]	KBS 2024	I3D	83.1	79.2	73.5	63.1	48	69.4
ADSFormer [11]	TMM 2024	I3D	84.4	80.0	73.1	62.9	46.9	69.5
		VideoMAE V2	85.3	80.8	73.9	64	49.8	70.8
BRTAL [30]	T-CSVT 2025	I3D	84.2	80.2	73.1	63	47.6	69.6
SAEFormer (Ours)	-	I3D	83.8	80.5	73.4	62.7	48.1	69.7
		VideoMAE	86.2	81.8	75.4	63.8	49.8	71.4

		V2	
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5.2.2 ActivityNet-1.3

On ActivityNet-1.3, SAEFormer achieves 36.5% average mAP with I3D features and 36.7% with TSP R(2+1)D features, as shown in Table 2. Compared with the corresponding I3D baseline, SAEFormer shows an improvement, whereas the gain is less pronounced under the TSP setting. This is partly because the ActivityNet localization protocol used here is class-agnostic, with only one foreground action class in the detection head. As a result, category-discriminative cues contribute less to the final metric, and performance depends more on proposal quality and long-range temporal coverage. Therefore, local boundary cue enhancement alone brings more limited gains on this benchmark.

Table 2.

Comparison with existing temporal action localization methods on the ActivityNet-1.3 dataset. The best results are highlighted in bold.

Method	Venue	Backbone	tIoU			Avg
			@0.5	@0.75	@0.95	
ActionFormer [10]	ECCV 2022	I3D	53.5	36.2	8.2	35.6
		R(2+1)D	54.7	37.8	8.4	36.6
TriDet [24]	CVPR 2023	R(2+1)D	54.7	38.0	8.4	36.8
TFFormer [33]	T-CSVT 2024	I3D	54.4	36.7	7.5	35.8
		R(2+1)D	54.5	37.3	7.6	36.2
ASL [28]	ICCV 2023	I3D	54.1	37.4	8.0	36.2
DualH [19]	FG 2024	I3D	54.4	37.1	8.1	36.3
FAM [46]	ICMRA 2024	I3D	54.6	37.2	8.3	36.3
MFAM [47]	TIP 2024	I3D	54.7	37.3	8.6	36.6
CTAN [26]	KBS 2024	R(2+1)D	55.6	38.6	7.6	37.2
GDD-TAD [27]	KBS 2024	R(2+1)D	56.5	38.6	8.5	37.4
BRTAL [30]	T-CSVT 2025	I3D	54.8	37.9	8.3	36.7
		R(2+1)D	55.1	38.1	8.5	37
SAEFormer	-	I3D	54.6	37.1	9.3	36.5

(Ours)		R(2+1)D	54.78	37.74	8.21	36.7
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5.2.3 EPIC-KITCHENS-100

On EPIC-KITCHENS-100, SAEFormer achieves 26.5% average mAP for verb localization and 23.7% for noun localization, as shown in Table 3. These results are consistent with the fine-grained and egocentric nature of EPIC-KITCHENS-100, where action instances are defined by verb-noun interactions in untrimmed first-person videos. Compared with activity-level benchmarks such as ActivityNet, verb localization in EPIC-KITCHENS-100 is more directly associated with hand motion dynamics, temporal transition patterns, and start-end boundary changes. These factors are aligned with the design of SAEFormer, which enhances local temporal cues and boundary-oriented representation cues. In contrast, noun localization depends more heavily on fine-grained object appearance and hand-object interactions. Since SAEFormer does not explicitly introduce object-centric modeling, the noun result should be interpreted as competitive rather than consistently superior to the strongest noun baseline.

Table 3.

Comparison with existing temporal action localization methods on the EPIC-KITCHENS-100 dataset. The best results are highlighted in bold.

V. (Verb)							
Method	Venue	tIoU					Avg
		@0.1	@0.2	@0.3	@0.4	@0.5	
ActionFormer [10]	ECCV 2022	26.6	25.4	24.2	22.3	19.1	23.5
TemporalMaxer [25]	arXiv 2023	27.8	26.6	25.3	23.1	19.9	24.5
ASL [28]	ICCV 2023	27.9	-	25.5	-	19.8	24.6
TriDet [24]	CVPR 2023	28.6	27.4	26.1	24.2	20.8	25.4

TE-TAD [29]	CVPR 2024	27.9	26.8	25.4	23.4	20	24.7
DualH [19]	FG 2024	28	27.1	25.4	23.1	19.9	24.7
GDD-TAD [27]	KBS 2024	29.5	28.2	26.9	24.1	20.8	25.9
BRTAL [30]	T-CSVT 2025	28.4	27.4	25.9	24.7	21	25.4
SAEFormer (Ours)	-	29.9	29.0	27.4	25.0	21.3	26.5
N. (Noun)							
Method	Venue	tIoU					Avg
		@0.1	@0.2	@0.3	@0.4	@0.5	
ActionFormer [10]	ECCV 2022	25.2	24.1	22.7	20.5	17	21.9
TemporalMaxer [25]	arXiv 2023	26.3	25.2	23.5	21.3	17.6	22.8
ASL [28]	ICCV 2023	26	-	23.4	-	17.7	22.6
TriDet [24]	CVPR 2023	27.4	26.3	24.6	22.2	18.3	23.8
DualH [19]	FG 2024	26.1	25	23.1	21.1	18	22.6
TE-TAD [29]	CVPR 2024	26.3	25.2	23.2	21	18.2	22.8
GDD-TAD [27]	KBS 2024	27.6	27.1	24.1	21	17	23.3
BRTAL [30]	T-CSVT 2025	27	25.7	23.7	21.5	18.2	23.2
SAEFormer (Ours)	-	27.7	26.5	24.4	21.8	18.2	23.7

5.3 Ablation Study

We conduct ablation studies on THUMOS14 to examine the contributions of the proposed components. Since the main manuscript focuses on the primary effectiveness of LMSF and BAE, we report the main component analysis under I3D features in Table 4. Additional ablations, including the VideoMAE V2 setting, internal variants of LMSF and BAE, and the transfer of BAE to ActionFormer, are provided in the supplementary material. Specifically, Tables S1-S4 report additional component and variant analyses, Table S5 provides diagnostic error analysis, Table S6 reports model complexity, and Figs. S1-S3 provide qualitative localization and visualization analyses.

As shown in Table 4, LMSF improves the average mAP from 67.89 to 69.16, and BAE alone improves it to 69.18. The full model further improves the average mAP to 69.70, achieving the best average performance among the listed variants. These results suggest that LMSF and BAE provide complementary effects in average localization performance. However, the improvements are not monotonically additive at every tIoU threshold. For example, LMSF alone obtains a slightly higher mAP at tIoU 0.5, while BAE alone obtains a higher mAP at tIoU 0.7. Therefore, we interpret the two components as complementary in terms of average localization performance rather than consistently superior at every individual threshold.

Table 4.

Main component analysis on THUMOS14 using I3D features. The best results are highlighted in bold.

Variant	LMSF	BAE	0.3	0.4	0.5	0.6	0.7	Avg
(A)	$\times$	$\times$	81.89	78.61	72.34	60.60	46.03	67.89
(B)	$\checkmark$	$\times$	83.57	80.10	73.53	61.74	46.84	69.16
(C)	$\times$	$\checkmark$	83.09	79.08	72.57	62.50	48.64	69.18
(D)	$\checkmark$	$\checkmark$	83.80	80.50	73.40	62.70	48.10	69.70

Note: (A) denotes the baseline without LMSF or BAE; (B) adds LMSF only; (C) adds BAE only; and (D) denotes the full SAEFormer configuration with both LMSF and BAE enabled.

6 Conclusion

We propose SAEFormer, a semantic-aware representation enhancement framework for temporal action localization. Instead of redesigning the detection pipeline, SAEFormer strengthens temporal representations through LMSF and BAE. LMSF preserves local

discriminative cues during stacked temporal attention, while BAE provides boundary-oriented residual reweighting without explicit boundary-level supervision.

The experiments show that SAEFormer is most effective in boundary-sensitive, multi-category action localization scenarios, where temporal transitions, action-background discrimination, and category-discriminative cues are important. Its gains are more limited in class-agnostic proposal-oriented localization and in object-centric noun localization, where performance depends more on long-range proposal quality, object appearance, and hand-object interactions. These results indicate that local temporal and boundary cue enhancement is useful but not sufficient for all TAL settings. Future work will explore adaptive cue weighting, object-aware representations, and more detailed boundary response analysis.

7 Data Availability

Data will be made available on request.

8 Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, we used AI-assisted tools for grammar checking, language polishing, manuscript organization, and experimental result data organization. The study design, scientific content, experimental results, figures, tables, and analyses were completed and verified by the authors. We did not use AI-assisted tools to generate,

fabricate, or alter research data, nor did we use generative AI to create, modify, or enhance images or figures in the submitted manuscript. After using these tools, we reviewed and edited the relevant content and take full responsibility for the content of the paper.

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